

Granularity and stability at coordinate interfaces

Static thresholds, planar response sets, and finite-rate instability

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Abstract

Positive first-order attachments to a single coordinate of a D-stable matrix have an exact distribution-free stability threshold: the first static D-instability along the corresponding positive diagonal ray. Below this threshold every finite positive partition is D-stable; at the threshold only a zero eigenvalue can obstruct stability; above it every sufficiently fine partition admits strict growth. This is an exact reduction to a static D-stability problem, not an arbitrary-dimensional decision procedure. We give a direct closed-right-half-plane absorption identity and a finite construction preserving an unstable eigenvalue. These yield quantitative partition bounds without a contour-limit argument. Planar response geometry also gives an exact test for a fixed partition. We extend safety to several coordinate ports and reciprocal relaxation modules, explain the relation to negative-imaginary systems, and show that four core states are necessary and sufficient for fixed-total granularity sensitivity. Exact rational examples distinguish stable and unstable two-piece partitions at the same total load. The single-port threshold theorem is verified end-to-end in Lean; the scope of the additional conventional proofs is stated in Section 9.

1 Problem and scope

A real square matrix is *Hurwitz* if all its eigenvalues have strictly negative real part, and *D-stable* if DB is Hurwitz for every positive diagonal D . Left and right scaling have the same spectra, since $DB = D(BD)D^{-1}$. We distinguish strict instability, meaning an eigenvalue in $\operatorname{Re} z > 0$, from a marginal eigenvalue on the imaginary axis.

Fix a D-stable $n \times n$ matrix B , a coordinate vector $e = e_p$, and a finite nonempty partition $r = (r_1, \dots, r_m)$ with $r_j > 0$. Write

$$A_r = \begin{pmatrix} B & er^T \\ \mathbf{1}e^T & -I_m \end{pmatrix}, \quad G = \sum_j r_j, \quad B_X = B + Xee^T. \quad (1)$$

Every core and attachment rate is free and positive. A scalar attachment with damping $h_j > 0$ and positive forward and return gains a_j, c_j has normalized load $r_j = a_j c_j / h_j$: a positive diagonal coordinate change and a reparameterization of its row rate give (1).

The question is how D-stability depends on the distribution of a fixed G . The answer retains the entire static loading interval. D-stability is not monotone under diagonal changes, and stable endpoints do not imply stability between them. Our results concern this restricted interface architecture; they do not settle the general characterization of D-stable matrices.

Schur complements, Routh–Hurwitz criteria, continuity of polynomial roots, and zero exclusion are classical ingredients [1, 2]. The dynamic-to-static reduction below uses the freedom of the rate at the *same coordinate* to which the attachments couple. That assumption is essential to the proof.

2 Direct spectral absorption

Inverse rates are convenient: z is an eigenvalue of DA precisely when $(zQ - A)v = 0$ for $Q = D^{-1} > 0$ and a nonzero vector v .

Lemma 2.1 (Closed-half-plane absorption). *Let $z = \sigma + i\omega \neq 0$, $\sigma \geq 0$, and $r, t > 0$. Set*

$$d = (1 + \sigma t)^2 + \omega^2 t^2, \quad \alpha = \frac{rt}{d}, \quad x = \frac{r(1 + 2\sigma t)}{d}. \quad (2)$$

Then $\alpha > 0$, $0 < x < r$, and

$$\frac{r}{1 + zt} + z\alpha = x. \quad (3)$$

Proof. The denominator is positive. Multiplication by the complex conjugate proves (3); moreover $r - x = r|z|^2 t^2 / d > 0$ and $1 + 2\sigma t > 0$. $\square$

Theorem 2.2 (Static-interval safety). *Suppose B_X is D -stable for every $0 < X < G$. Then every positive partition of G has no nonzero eigenvalue in the closed right half-plane under any positive scaling. It is D -stable if and only if $\det B_G \neq 0$. If $\det B_G = 0$, its only possible eigenvalues outside the strict left half-plane are zero eigenvalues.*

Proof. For a nonzero candidate z with $\operatorname{Re} z \geq 0$, every $1 + zt_j$ is nonzero. Eliminating the leaves gives

$$\left(zQ - B - ee^T \sum_j \frac{r_j}{1 + zt_j} \right) v = 0.$$

The core vector v is nonzero, since a zero core vector would force every leaf coordinate to vanish. Apply Lemma 2.1 termwise. With $X = \sum_j x_j \in (0, G)$ and $Q' = Q + (\sum_j \alpha_j) ee^T > 0$, the same equation is $(zQ' - B_X)v = 0$, contradicting D -stability of B_X . At $z = 0$, ordinary block elimination gives $\det(-A_r) = \det(-B_G)$. Positive scaling multiplies the determinant by a positive factor, so occurrence of a zero eigenvalue is independent of the partition and rates. $\square$

This proof includes real positive eigenvalues and does not require spectral continuation or stability of the principal block obtained by deleting p . At $\sigma = 0$ it is the familiar phase absorption $X = \sum r_j / (1 + t_j^2)$ and $q'_p = q_p + \sum r_j t_j / (1 + t_j^2)$.

3 The exact threshold and finite realization

Define

$$\tau = \inf\{X \geq 0 : B_X \text{ has a strictly unstable positive scaling}\}. \quad (4)$$

The infimum is finite: D -stability implies $B_{pp} \leq 0$, while $X > -B_{pp}$ makes the port diagonal positive. Increasing its positive column scaling makes the trace positive, forcing an eigenvalue with positive real part. Thus $0 \leq \tau \leq -B_{pp}$.

Lemma 3.1 (Static marginality below first strict growth). *B_X is D -stable for $0 \leq X < \tau$.*

Proof. Strict growth is excluded by definition, and $X = 0$ is covered by the hypothesis on B . Suppose a positive load X has a contact $i\omega$ at inverse rates $Q > 0$. Put $P(z) = zQ - B$ and $c(z) = \operatorname{adj}(P(z))_{pp}$. Affineness of the determinant gives $\det(P(i\omega) - Xee^T) = \det P(i\omega) - Xc(i\omega) = 0$.

The unloaded determinant is nonzero by D-stability, hence $c(i\omega) \neq 0$. The continuous quotient $f(z) = \det P(z)/c(z)$ is therefore defined near the contact and satisfies $f(i\omega) = X$.

For small $\sigma > 0$, set $z = \sigma + i\omega$ and, if $\omega \neq 0$, put

$$k(\sigma) = -\frac{\operatorname{Im} f(z)}{\omega}, \quad Y(\sigma) = \operatorname{Re} f(z) + \sigma k(\sigma).$$

If $\omega = 0$, use $k(\sigma) = 0$ and $Y(\sigma) = f(\sigma)$, which is real. In either case $k(\sigma) \rightarrow 0$, $Y(\sigma) \rightarrow X$, and $f(z) + zk(\sigma) - Y(\sigma) = 0$. Changing just the port inverse rate to $q_p + k(\sigma)$ and the load to $Y(\sigma)$ thus makes the determinant zero at z , while preserving positive rates and positive load for small σ . This produces strict growth arbitrarily close to X , and therefore $\tau \leq X$, a contradiction. $\square$

Lemma 3.2 (Finite realization of static growth). *Suppose $0 < X < G$ and B_X has an eigenvalue $z = \sigma + i\omega$, $\sigma > 0$, at positive inverse core rates Q . There is $\varepsilon > 0$ such that every partition of G with $\max_j r_j < \varepsilon$ realizes exactly this eigenvalue at finite positive rates.*

Proof. Let $q = Q_{pp}$, $k = |z|^2$, and

$$f(t) = \frac{1 + 2\sigma t}{1 + 2\sigma t + kt^2}.$$

This function decreases continuously from 1 to 0 on $(0, \infty)$: $f'(t) = -2kt(1 + \sigma t)/(1 + 2\sigma t + kt^2)^2 < 0$. Choose $\delta > 0$ satisfying

$$G\delta + \frac{Gk\delta^2}{2\sigma} < q, \quad Gk\delta^2 < G - X, \quad (5)$$

and choose $0 < \varepsilon < \min\{G - X, X[1 - f(\delta)]\}$. The first partial sum S strictly greater than X obeys $X < S < X + \varepsilon < G$. Give these leaves inverse rate δ . The remaining leaves receive a common inverse rate T determined by

$$f(T) = y := \frac{X - Sf(\delta)}{G - S}. \quad (6)$$

Here $y > 0$ follows from $S - X < X[1 - f(\delta)]$. Also $S[1 - f(\delta)] \leq Gk\delta^2 < G - X$, which proves $y < 1$. Thus a unique finite $T > 0$ exists; explicitly

$$T = \frac{\sigma(1 - y) + \sqrt{\sigma^2(1 - y)^2 + ky(1 - y)}}{ky}.$$

The total effective static load in (2) is exactly X . The fast leaves contribute at most $G\delta$ to $\sum \alpha_j$. For each slow leaf, $\alpha_j \leq x_j/(2\sigma)$, so their contribution is at most $X[1 - f(\delta)]/(2\sigma) \leq Gk\delta^2/(2\sigma)$. By (5) the total is strictly below q . Replace the core hub inverse rate by $q - \sum \alpha_j > 0$, retain every other core inverse rate, and reverse (3). Reconstructing the leaf eigenvector gives a nonzero eigenvector of the attached matrix at precisely z . All rates are finite. $\square$

Theorem 3.3 (Exact distribution-free granularity threshold). *For (1) and (4):*

1. *If $0 < G < \tau$, every positive partition is D-stable.*
2. *If $G = \tau > 0$, every partition is D-stable exactly when $\det B_\tau \neq 0$. Otherwise only zero eigenvalues can accompany its strictly left-half-plane spectrum.*
3. *If $G > \tau$, every sufficiently fine partition has a strictly unstable positive scaling.*

Proof. Lemma 3.1 and Theorem 2.2 prove the first two claims; below τ the determinant is nonzero by the same lemma. For $G > \tau$, the defining infimum supplies a strictly unstable static load $X \in (0, G)$. Apply Lemma 3.2. $\square$

The quantity τ is explicit only when the static problem can be solved. The theorem is a reduction, not a claim that evaluating the infimum is easy. Equations (5)–(6) give a constructive fineness bound from any known static witness, without optimizing its rate spread.

4 Planar geometry and fixed partitions

4.1 A hub connecting stable modules

Consider a hub of self-coefficient b and modules M_j with input w_j and return c_j^T . Assume every M_j is D-stable. Define

$$V_j = \{c_j^T(iQ_j - M_j)^{-1}w_j : Q_j > 0 \text{ diagonal}\}, \quad G_j = c_j^T(-M_j)^{-1}w_j.$$

Frequency normalization is legitimate because all positive diagonal rates are free. It does not assert that V_j is closed under complex scalar multiplication.

Theorem 4.1 (Hub value-set criterion). *The full hub-and-module matrix is D-stable if and only if*

$$-b - \sum_j G_j > 0, \quad \left(\sum_j V_j \right) \cap \{-b + iq : q > 0\} = \emptyset.$$

Proof. Every module pencil on the imaginary axis is invertible. Schur elimination at frequency one gives $iq_h - b - \sum_j H_j = 0$. Since $q_h > 0$ is free, this is exactly intersection with the stated ray. At zero, $\det(-A) = \prod_j \det(-M_j)(-b - \sum_j G_j)$. Necessity follows from Hurwitz stability and positivity of these determinants.

For sufficiency, fix positive module rates and let the hub rate tend to zero. At zero hub rate the modules have strictly stable eigenvalues and there is one simple zero eigenvalue. The derivative of that eigenvalue with respect to hub rate is $b + \sum_j G_j < 0$, as follows by differentiating the scalar Schur equation and the simple-root implicit-function theorem. Thus a Hurwitz scaling exists. The positive diagonal cone is path connected. Along any compact path, eigenvalues are bounded and continuous as a multiset; leaving the Hurwitz region requires a zero or nonzero imaginary root. Both have been excluded. $\square$

In particular, if every V_j lies in the closed lower half-plane, the positive DC margin alone is necessary and sufficient. An oscillatory loss through this architecture requires at least one module capable of positive imaginary response. The positive DC margin supplies the base point even for nonreciprocal D-stable modules; reciprocity is unnecessary for this particular step.

4.2 The exact protection envelope

For scalar positive leaves, write

$$\sum_j \frac{r_j}{1 + it_j} = X - iY.$$

For $0 < X < G$ define

$$\Lambda_r(X) = \min_{\substack{1 \leq k \leq m, \\ S \subseteq \{1, \dots, m\} \setminus \{k\} \\ r(S) \leq X \leq r(S) + r_k}} \sqrt{(X - r(S))(r(S) + r_k - X)}. \quad (7)$$

The notation in the minimum means that k is an index and S is a subset of the remaining indices; $r(S) = \sum_{j \in S} r_j$.

Lemma 4.2 (Exact leaf image). *For one leaf, $Y = \sqrt{X(G-X)}$. For at least two leaves the finite-rate heights at fixed X are exactly*

$$\Lambda_r(X) < Y \leq \sqrt{X(G-X)}.$$

Proof. One leaf has $x = r/(1+t^2)$ and $y = \sqrt{x(r-x)}$. On the slice $0 \leq x_j \leq r_j$, $\sum x_j = X$, the sum of these strictly concave heights has its maximum at $x_j = r_j X/G$, by concavity and homogeneity. Its minimum is achieved at a polytope vertex, where at most one coordinate is not an endpoint. These vertices give (7). With at least two leaves a point with every coordinate interior cannot minimize a strictly concave function. The segment from a minimizing vertex to the synchronized maximum lies in the interior except at its initial point and realizes all intermediate heights. Finally $t_j = y_j/x_j > 0$ reconstructs finite rates. $\square$

For a D-stable core $B = \begin{pmatrix} b & a^T \\ c & C \end{pmatrix}$, put $Z(Q_c) = a^T(iQ_c - C)^{-1}c$ wherever the inverse exists, and define the extended envelope

$$Q^*(X) = \sup\{\text{Im } Z(Q_c) : Q_c > 0, \text{Re } Z(Q_c) = -b - X\}, \quad \sup \emptyset = -\infty.$$

Theorem 4.3 (Exact fixed-partition test). *A_r is D-stable if and only if*

$$\frac{\det(-B_G)}{\det(-B)} > 0, \quad Q^*(X) \leq \Lambda_r(X) \quad (0 < X < G). \quad (8)$$

For one leaf use $\Lambda_r(X) = \sqrt{X(G-X)}$.

Proof. An imaginary contact requires $q_h = \text{Im } Z - Y > 0$ and $X = -b - \text{Re } Z$. Lemma 4.2 shows that such a contact exists exactly when $Q^*(X) > \Lambda_r(X)$ for some X . The strict inequality is essential: $q_h = 0$ is not an admissible rate, and for multiple leaves the lower height is only a limiting value.

Singular inner pencils lose no contacts in this argument. If $d = \det(iQ_c - C) = 0$, put $n = a^T \text{adj}(iQ_c - C)c$. The core determinant equals $(iq_h - b)d - n = -n$, hence $n \neq 0$ because B is D-stable. Adding leaves changes only the coefficient of d and cannot make the determinant vanish.

For a stable base point choose fixed positive core rates and give all leaves rate ϵ . The fast eigenvalues converge to those of the stable core. Schur elimination with $z = \epsilon\mu$ shows that the slow eigenvalues, divided by ϵ , converge to those of $-I + k\mathbf{1}r^T$, where $k = e^T(-B)^{-1}e$. They are -1 and $-1 + kG$. The determinant condition is $1 - kG > 0$, so small positive ϵ gives a Hurwitz base point. Zero exclusion along the positive rate cone proves sufficiency. Necessity of the determinant sign and of contact exclusion is immediate. $\square$

The exact test may require exponentially many subset labels and a difficult core-envelope computation. Planar geometry does not remove those costs. For negative-load leaves the maximum positive imaginary height at fixed negative real response is attained with synchronized times and depends only on total absolute load. The same contact and base-point reasoning therefore makes their all-rate classification independent of partition. Positive leaves differ because their *minimum* protective height is partition dependent.

5 Several ports and reciprocal attachments

Let P be a set of distinct coordinates, each with positive total G_p , and put $B_X = B + \sum_{p \in P} X_p e_p e_p^T$ and $\mathcal{U} = \prod_{p \in P} (0, G_p)$.

Theorem 5.1 (Open-box criterion). *Every choice of positive finite partitions at the ports is D-stable if and only if every B_X , $X \in \mathcal{U}$, has no strictly unstable positive scaling and $\det B_G \neq 0$.*

Proof. First, absence of strict growth on the open box implies D-stability there. For fixed $Q > 0$, $\det(iQ - B_X)$ is multiaffine in X . At a zero, a nonzero coordinate partial allows the one-coordinate perturbation in Lemma 3.1, producing nearby strict growth inside the box. Thus all coordinate partials vanish at every zero. Affineness propagates the zero along each coordinate line in the box; the argument is reapplied at every point reached. Iterating through the coordinates makes the determinant identically zero, contradicting its limiting value at $X = 0$. For zero frequency, $\det(-B_X) \geq 0$ under no strict growth: a negative constant term would produce a positive real root. A nonnegative affine function vanishing at an interior point has zero slope. The same propagation contradicts $\det(-B) > 0$.

Apply Lemma 2.1 independently at all ports. Every nonzero closed-right-half-plane candidate maps into $\mathcal{U}$, where it is excluded. The total-load determinant handles zero.

Conversely, strict growth at an interior X is realized by sufficiently fine partitions: use Lemma 3.2 independently at every port, preserving the same eigenvalue and subtracting the respective inverse-rate shifts. If the total-load determinant vanishes, every partition is singular. Strict growth on a face also persists at nearby interior points by openness, but no assumption of stability on all faces is required. $\square$

Corollary 5.2 (Reciprocal relaxation safety). *A module with dynamics $\dot{z} = D_z(Mz + wx_p)$ and return $\alpha w^T z$, where $M = M^T < 0$ and $\alpha > 0$, obeys the same safety conclusions with total load $G = \alpha w^T (-M)^{-1} w$. Multiple such modules can be combined at one or several coordinate ports.*

Proof. Let $S = D_z^{1/2} M D_z^{1/2} < 0$ and $u = D_z^{1/2} w$. The spectral theorem gives

$$H(s) = \alpha u^T (sI - S)^{-1} u = \sum_{\ell} \frac{a_{\ell}}{s + \lambda_{\ell}} = \sum_{\ell} \frac{g_{\ell}}{1 + s/\lambda_{\ell}}, \quad a_{\ell} \geq 0, \quad \lambda_{\ell} > 0,$$

where $\sum g_{\ell} = H(0) = G$ independently of D_z . Omit zero residues and apply the scalar absorption argument at each fixed scaling. The internal poles are strictly negative. $\square$

For a positive diagonal symmetrizer H_0 with $H_0 M$ symmetric negative definite, the reciprocal output must be $\alpha w^T H_0$. Conservation zero modes require separate treatment. The corollary is a safety result; its converse does not permit independently tuning the modal rates of a fixed indivisible subnetwork.

These are classical relaxation transfer functions in the sense of Willems [4, 5]. Lanzon and Petersen's negative-imaginary theorem [3] likewise decides an appropriate feedback interconnection by its DC loop gain. Here the arbitrary core transfer is not assumed to be negative imaginary for every diagonal scaling. Coordinate-rate absorption, and the *first* static loss rather than a single endpoint DC test, are what allow the present threshold theorem to retain static reentry.

6 Small cores and the minimum dimension

For a two-state D-stable core $B = \begin{pmatrix} -a & p \\ q & -b \end{pmatrix}$, let $t = ab - pq > 0$. Then

$$\tau = \begin{cases} \min\{a, t/b\}, & b > 0, \\ a, & b = 0. \end{cases}$$

At positive threshold equality is safe precisely when $t - b\tau > 0$.

For three states define signed principal-minor data

$$a = -B_{11}, \quad b = -B_{22}, \quad c = -B_{33}, \quad u = \det B_{12}, \quad v = \det B_{13}, \quad w = \det B_{23}, \quad t = \det(-B) > 0.$$

These are nonnegative except that t is strictly positive. For loading coordinate one put

$$L = \min(\{a\} \cup \{u/b : b > 0\} \cup \{v/c : c > 0\} \cup \{t/w : w > 0\}),$$

and, on $[0, L]$, set $A = w(a - X)$, $E_2 = b(v - cX)$, $E_3 = c(u - bX)$, and $T = t - wX$. The classical cubic principal-minor criterion for absence of strict growth is

$$T \leq (\sqrt{A} + \sqrt{E_2} + \sqrt{E_3})^2. \quad (9)$$

For completeness, the rate-dependent cubic Hurwitz inequality is $(\sum a_i d_i)(\sum u_{ij} d_i d_j) \geq t d_1 d_2 d_3$. After division by the positive product, its infimum is the squared sum in (9), by Cauchy–Schwarz; zero coefficients are obtained by limits. A strict violation is realized at finite rates. This is the usual small-dimensional criterion, not a new general D-stability test.

Proposition 6.1 (No reentry for at most three core states). *Along a positive coordinate loading ray from a D-stable core of order at most three, strict instability cannot be followed by recovery of absence of strict growth. Consequently granularity sensitivity at fixed total load requires at least four core states.*

Proof. In three dimensions the gap between the right and left sides of (9) is a sum of an affine function and terms $2\sqrt{UV}$ with U, V nonnegative affine functions of X . The geometric mean is concave on the nonnegative quadrant, so this gap is concave on $[0, L]$. Its nonnegative set is an interval containing zero. Beyond L a nonincreasing principal minor is negative, forcing strict growth at a suitable scaling. The two-state assertion is immediate from its formula. Thus above τ the total-load static matrix is strictly unstable. For every partition sufficiently fast leaves approximate this unstable static matrix, and strict growth persists at finite rates. Below and at τ Theorem 3.3 is independent of partition. $\square$

An explicit algebraic threshold prescription is available in three states. Let $E = T - A - E_2 - E_3$ and $V = AE_2 + AE_3 + E_2 E_3$. The polynomial

$$P = E^4 - 8VE^2 - 64AE_2 E_3 E + 16\{V^2 - 4AE_2 E_3(A + E_2 + E_3)\}$$

has degree at most two in X for these affine data. It is the product of $T - (\sqrt{A} \pm \sqrt{E_2} \pm \sqrt{E_3})^2$ over the four sign choices. If it is identically zero, $\tau = L$. Otherwise isolate its roots in $(0, L)$ and test (9) between consecutive roots. The first violating interval starts at τ , or $\tau = L$ if none exists. The original radical test rejects extraneous squared branches; concavity ensures there is only one transition to failure. Equality of the attached system is decided by $t - w\tau$.

7 An exact four-state contrast

Consider

$$B(h) = \begin{pmatrix} -1 & -7/10 & -7/10 & -14/5 \\ 4 & -1 & -7/10 & -14/5 \\ 4 & 4 & -1 & -14/5 \\ -1 & -1 & -1 & -h \end{pmatrix}, \quad h \geq 4.$$

With $K(h) = 3890h^2 - 38717h + 93492$ and $h_{\pm} = (38717 \pm 329\sqrt{409})/7780$, its exact classification is

$$B(h) \text{ D-stable} \iff 4 \leq h \leq h_- \text{ or } h \geq h_+. \quad (10)$$

Here is a compact certificate. For core rates x, y, z and hub rate v set $p = x + y + z$, $q = xy + xz + yz$, $r = xyz$, $U = pq - 9r \geq 0$ and $V = q^2 - 3pr \geq 0$. The quartic coefficients are

$$\begin{aligned} c_1 &= p + hv, & c_2 &= \frac{19}{5}q + (h - \frac{14}{5})vp, \\ c_3 &= \frac{466}{25}r + (\frac{19}{5}h - \frac{371}{25})vq, & c_4 &= (\frac{466}{25}h - \frac{18613}{250})vr. \end{aligned}$$

They are positive for $h \geq 4$. Writing $c_1c_2c_3 - c_3^2 - c_1^2c_4 = \sum_{j=0}^3 F_jv^j$, direct expansion gives

$$\begin{aligned} F_0 &= \frac{466}{625}r(95U + 389r), \\ 1250F_1 &= (18050h - 70490)qU + (73910h + 57134)qr + 27825p^2r, \\ 625F_2 &= 5(5h - 14)(95h - 371)pU + 371(95h - 371)V \\ &\quad + (9725h^2 + 22855h - 179193)pr, \\ 250F_3 &= h\{2(5h - 14)(95h - 371)U + K(h)r\}. \end{aligned}$$

All terms except the coefficient $K(h)$ are positive or nonnegative. If $K(h) \geq 0$, Routh–Hurwitz proves stability at every finite rate. If $K(h) < 0$, choose $x = y = z = 1$ and then large v ; the last expression has negative leading coefficient. This proves (10), including strict stability at both endpoints.

At port four of $B(6)$, $\tau = 6 - h_+ = (7963 - 329\sqrt{409})/7780$ and threshold equality is safe. At total $G = 2$ the following exact verdicts hold.

Partition	Verdict	Certificate
(2)	D-stable	quintic Hurwitz determinants
(19/10, 1/10)	D-stable	four terminal edges and corners
(1, 1)	strictly unstable scaling exists	rational Routh witness
(1/2, 1/2, 1/2, 1/2)	strictly unstable scaling exists	refinement of (1, 1)

Refinement preserves an existing witness by assigning equal rates to leaves within each old group; the additional difference modes are strictly stable.

For (1, 1) a finite witness is

$$D = \text{diag}(617/1000, 617/1000, 617/1000, 35, 158/5, 4/125).$$

The characteristic polynomial has coefficients, in descending order,

$$\begin{aligned} &1, \quad \frac{243483}{1000}, \quad \frac{29034069433}{5000000}, \quad \frac{63896127507669}{12500000000}, \\ &\frac{36179911928121089}{625000000000}, \quad \frac{76473766154601901}{15625000000000}, \quad \frac{3507069622203}{3906250000000}. \end{aligned}$$

Its exact Routh first-column signs are $+, +, +, +, -, +, +$, with no zero pivots: there are two strictly right-half-plane roots. All coefficients are positive, so the unstable roots are nonreal. The achieved rate spread is $\max D_{ii} / \min D_{ii} = 4375/4 = 1093.75$; no optimality is claimed.

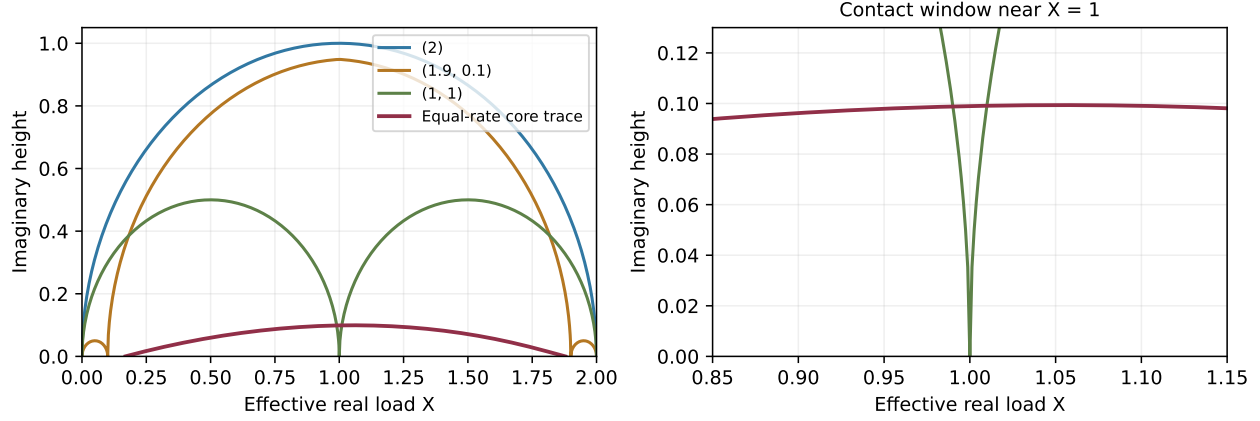


Figure 1: Partition protection envelopes and an illustrative equal-inner-rate core trace for $B(6)$. The right panel enlarges the contact near $X = 1$. The red trace is a subset of the core value set, not a certified plot of the supremum Q^* . The stable verdicts use exact edge certificates rather than an inference from this sampled trace.

For reproducibility of the stable verdicts, the terminal single-leaf matrix with damping h and gain g has coefficients

$$\begin{aligned} &1, \quad c_1 + w, \quad c_2 + w[p + (h - g)v], \\ &c_3 + w[\frac{19}{5}q + (h - g - \frac{14}{5})vp], \\ &c_4 + w[\frac{466}{25}r + (\frac{19}{5}(h - g) - \frac{371}{25})vq], \\ &w[\frac{466}{25}(h - g) - \frac{18613}{250}]vr. \end{aligned}$$

For $(h, g) = (6, 2)$ and each of the four pairs

$$(6, 19/10), \quad (59/10, 19/10), \quad (6, 1/10), \quad (41/10, 1/10),$$

form $\Delta_k = \det(a_{2j-i+1})_{i,j=0}^{k-1}$ for $k = 1, \dots, 4$, with $a_0 = 1$ and missing coefficients zero. Substitution $x = a + b + c$, $y = b + c$, $z = c$ gives only nonnegative rational coefficients, and a positive polynomial at $a = b = 0$. Permutation symmetry of p, q, r covers every ordering of the inner rates. The fifth determinant is $a_5 \Delta_4 > 0$.

The four edges suffice for the two-leaf verdict: a violating response lies strictly above a minimizing semicircle in (7). An interior point of that semicircle gives a contact for its one-leaf edge, with the other leaves fixed at their static endpoint loads. At a semicircle endpoint it gives an imaginary contact for the corresponding static corner. All these edges and corners are D-stable here. The full determinant is that of $B(4)$ and has positive sign. Theorem 4.3 applies. Together with Proposition 6.1, this establishes the four-state minimum for the fixed-core question.

8 Buffers, decoys, and what the theorem predicts

A linear reversible site with binding coefficient $k > 0$, unbinding $u > 0$, and bound-complex degradation $\delta \geq 0$ contributes a port loss k and positive load

$$r = \frac{ku}{u + \delta}.$$

For a fixed linearized core the relevant static loading interval is therefore the reachable damping interval from the loss-only core to the fast-site Schur complement. If both endpoints are D-stable but its interior contains strict growth, sufficiently fine heterogeneous site classes reveal that hidden instability. In general the criterion is reachable static growth; two stable endpoints, and hence literal reentry, are extra hypotheses.

For example, adding total binding loss two to $B(4)$ produces $B(6)$. With no bound degradation the return total is two. The preceding example distinguishes one site class, two unequal classes, and two equal classes under independent positive rates. This is an exact linear interface example, not a calibrated biochemical clock.

Decoy-induced creation and suppression of oscillations are already known in nonlinear biomolecular-clock models [6, 7]; they are not novel phenomena claimed here. Nonlinear site addition can change the equilibrium and the core Jacobian, whereas our comparison holds that Jacobian fixed. In particular a fixed three-state Goodwin core cannot serve as the four-state granularity example. The conversion from row-rate freedom to physically linked kinetic parameters must also be checked in each application.

The maximum of Q^* alone cannot bound the full rate spread needed for instability: it omits the inner rates attaining that height. A quantitative statement must retain a core response witness and its rates. The construction in Lemma 3.2 does so. The rational example above gives an achieved spread, not a universal “fragility index” or a minimum-spread certificate.

9 Verification and limits

Theorem 3.3, including $0 \leq \tau \leq -B_{pp}$ and its complete equality case, is verified end-to-end in Lean against the frozen mathlib environment. The exported theorem is `granularity_threshold` in the namespace `DStabilityCharacterization.Granularity`, defined in `Threshold.lean`. It uses literal matrix eigenpairs under every positive diagonal scaling, for arbitrary finite core and leaf index sets. The tolerance in the converse is quantified before the leaf set and partition.

The proof imports the actual matrix absorption and zero-endpoint theorems, the contact perturbation of Lemma 3.1, the uniform finite realization of Lemma 3.2, and the trace argument bounding the threshold. No spectral-continuation, coverage or realization premise is assumed. Strict compilation treats warnings as errors, contains no proof placeholders, and the exported theorem depends only on Lean’s standard axioms `propext`, `Classical.choice`, and `Quot.sound`.

The additional hub, fixed-partition, multiport, reciprocal and minimum-dimension results have conventional proofs here. Their full formal assembly, including multiaffine propagation, positive-residue decomposition, and envelope geometry, is not claimed. Exact rational symbolic replay checks the examples and polynomial certificates; their generic fifth- and sixth-order Routh implications are not represented as Lean spectral theorems.

The exact threshold and envelope are reductions. General static D-stability, computable envelopes for arbitrary interiors, and non-coordinate interfaces remain outside the scope. The numerical examples establish linear spectral growth; they do not prove a nonlinear Hopf bifurcation or a stable limit cycle.

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